UNIT 6 REVIEW · STUDENT WORKSHEET

Functions, Linear Equations & *Systems*

Name: _____ **Date:** _____ **Period:** _____

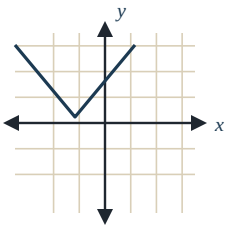
25 problems covering Sections 6.1, 6.2, 6.3, and 6.7.

Quick Reference Unit 6 Key Formulas & Concepts

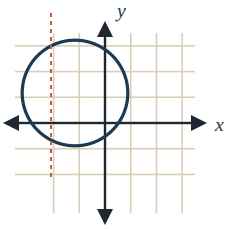
Function (6.1)

A relation in which every **input (x)** has exactly **one output (y)**.

Vertical Line Test: if any vertical line crosses the graph more than once, it's *not* a function.



✓ Function



✗ Not a function

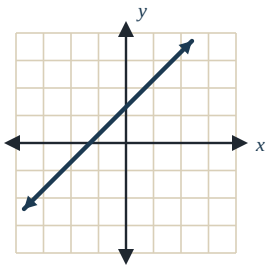
Function Notation (6.1)

`f(x) = 2x + 4` reads "*f of x*".

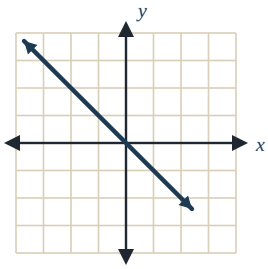
To **evaluate**, substitute the input for x using parentheses.
Example: f(3) = 2(**3**) + 4 = **10**.

Domain = all x-values (inputs)
Range = all y-values (outputs)

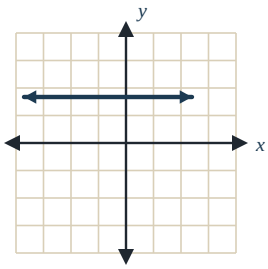
Four Types of Slope (6.2)



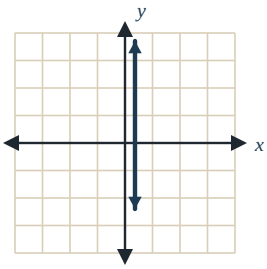
Positive
Up left to right
m > 0



Negative
Down left to right
m < 0



Zero
Horizontal
m = 0



Undefined
Vertical
m = undefined

Slope-Intercept Form (6.2)

$$y = mx + b$$

m = slope (rate of change; the number multiplied by x)
b = y-intercept (where the line crosses the y-axis; constant)

Slope formula: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$

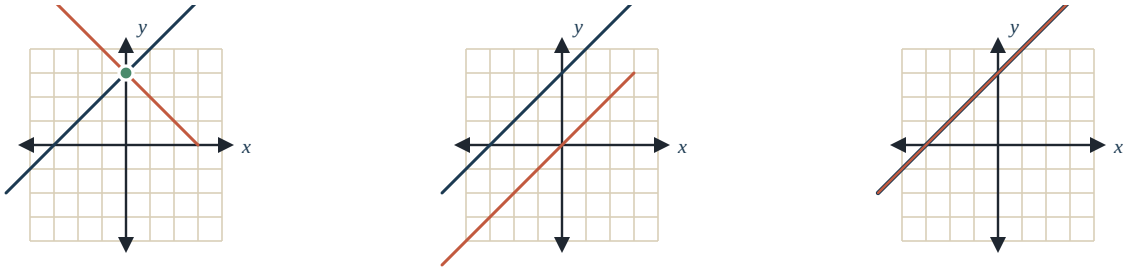
Table → Equation (6.3)

Linear if `Δy ÷ Δx` is the *same* between every pair of rows.

Steps: (1) Find slope $m = (y_2 - y_1) \div (x_2 - x_1)$. (2) Find b: use the row where x = 0, or solve y = mx + b for b. (3) Write y = mx + b.

Systems of Equations (6.7)

A **solution** is the (x, y) point that makes *both* equations true — graphically, where the lines intersect.



One Solution
Different slopes

No Solution
Same slope, different b

Infinite Solutions
Same slope, same b

Substitution: Solve one equation for a variable, substitute that expression into the other equation, solve, then plug back in to find the second variable.

6.1 Introduction to Functions

1. `VOCABULARY`

Which statement **best defines** a function?

A. A function is any equation that contains an x and a y.

B. A function is a relation in which every input has exactly one output.

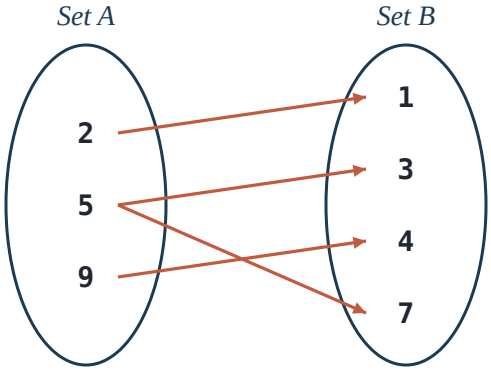
C. A function is a relation in which every output has exactly one input.

D. A function is the graph of a straight line.

---

2. `MAPPING DIAGRAM`

Does the mapping diagram below represent a function? Why or why not?



A. Yes — every input is paired with an output.

B. No — the input `5` is paired with two different outputs (`3` and `7`).

C. Yes — no two outputs are repeated.

D. No — the output `7` is paired with only one input.

---

3. `ORDERED PAIRS`

Which set of ordered pairs does *NOT* represent a function?

A. `{(1, 2), (3, 4), (5, 6), (7, 8)}`

B. `{(−2, 5), (0, 5), (4, 5), (8, 5)}`

C. `{(3, 1), (3, 2), (5, 4), (6, 7)}`

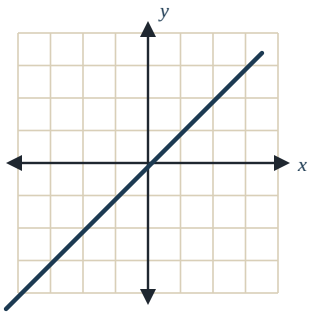
D. `{(−1, 0), (0, −1), (1, 1), (2, 3)}`

---

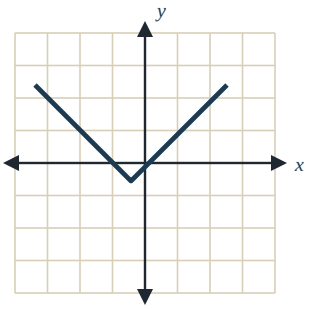
4. `VERTICAL LINE TEST`

Which graph does *NOT* represent a function? (Apply the vertical line test.)

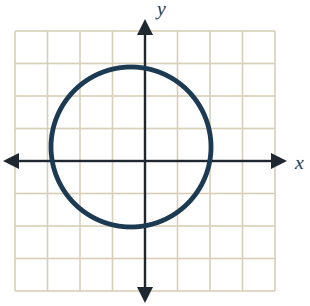
A



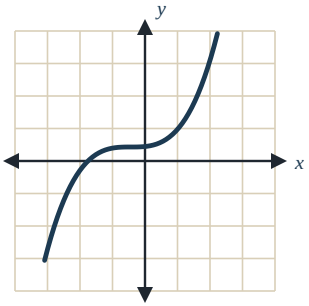
B



C



D



---

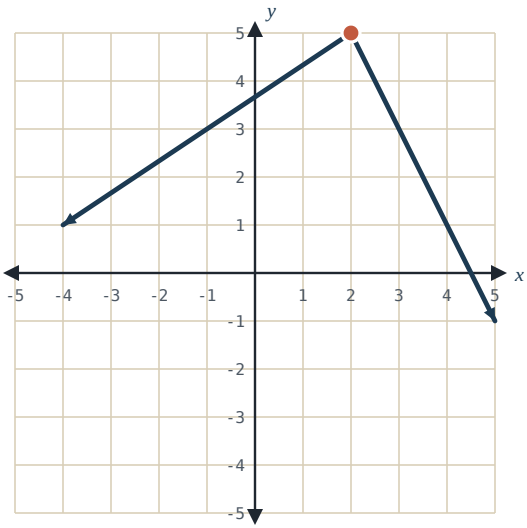
5. `FUNCTION NOTATION`

Given `f(x) = 2x − 7`, find `f(−3)`.

`f(−3) =` _____

6. `EVALUATE FROM GRAPH`

Use the graph of `y = f(x)` shown below to find `f(2)`.



`f(2) =` _____

7. `DOMAIN & RANGE`

Find the **domain** and **range** of the function defined by the table below. Express each as a set in `{ }` braces.

x	−3	0	5	8
y	4	−2	7	−2

`Domain =` _____

`Range =` _____

8. VOCABULARY

For each line description, identify the type of slope.

A horizontal line _____

A vertical line _____

A line going up from left to right _____

A line going down from left to right _____

Choose from: Positive · Negative · Zero · Undefined

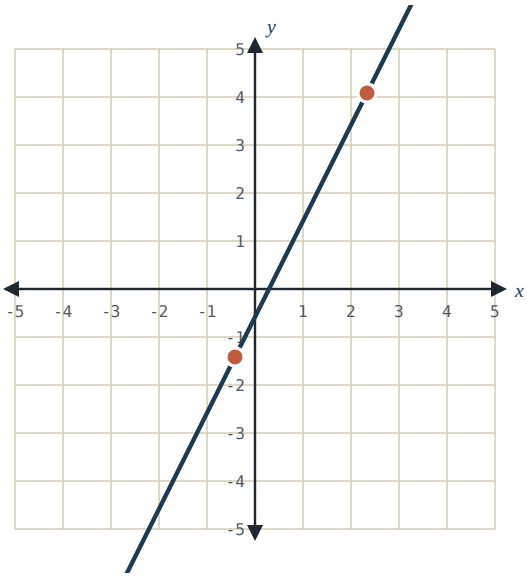
9. SLOPE FROM TWO POINTS

Find the slope of the line through `(−2, 5)` and `(4, −1)`. Write your answer in simplest form.

m = _____

10. SLOPE FROM GRAPH

Find the slope of the line shown on the graph below.



m = _____

11. IDENTIFY SLOPE

Find the **slope** of the line: $y = -\frac{3}{4}x + 5$

m = _____

12. IDENTIFY Y-INTERCEPT

Find the **y-intercept** of the line: $y = 2x - 9$

b = _____

13. LINEAR OR NOT

Which table represents a **linear function**?

A

x	0	1	2	3
y	1	2	4	8

B

x	-2	-1	0	1
y	4	1	0	1

C

x	0	1	2	3
y	7	4	1	-2

D

x	0	1	2	3
y	0	1	4	9

14. ERROR ANALYSIS

A student tried to find the slope of the line through `(2, −3)` and `(8, 9)`. Their work is shown below. Identify the error and find the **correct slope**.

STUDENT'S WORK

m = (8 − 2) ÷ (9 − (−3))

m = 6 ÷ 12

m = **1/2**

Correct m = _____

15. `SOLUTIONS TO A LINEAR EQUATION`

Which of the following is a **solution** to the linear equation `y = −2x + 5`?

A. `(1, 2)`

B. `(3, −1)`

C. `(0, −5)`

D. `(−2, 1)`

6.3 Writing Equations from Tables

16. `MISSING VALUE`

Assuming a linear relationship, find the **missing value** in the table below.

x	1	2	3	4	5
y	7	11	15	19	?

y = _____

17. `IDENTIFY LINEAR TABLE`

Which table represents a **linear function**?

A

x	1	2	3	4
y	2	6	12	20

B

x	-3	-1	1	3
y	8	4	0	-4

C

x	0	1	2	3
y	5	6	8	11

D

x	0	2	4	6
y	1	3	7	13

18. `EQUATION FROM TABLE (X = 0)`

Find the equation of the linear function in **slope-intercept form**.

x	0	1	2	3	4
y	−5	−2	1	4	7

y = _____

19. `EQUATION FROM TABLE (NO X = 0)`

Find the equation of the linear function in **slope-intercept form**. Note: the table does not include the x = 0 row.

x	1	2	3	4
y	1	−3	−7	−11

y = _____

20. `ERROR ANALYSIS`

A student wrote an equation for the linear table below. Identify the error and write the **correct equation**.

x	0	1	2	3
y	4	7	10	13

STUDENT'S WORK

Student's answer: **y = 4x + 3**

Correct equation: y = _____

6.7 Systems of Equations

21. `VOCABULARY`

A **solution** to a system of two linear equations is best described as…

A. any point that lies on either line.

B. the point where both lines have the same slope.

C. the `(x, y)` coordinate that makes *both* equations true.

D. the x-value where one line equals zero.

22. `SOLUTIONS BY INSPECTION`

How many solutions does this system have? `y = 2x + 5` `y = 2x − 1`

A. One solution

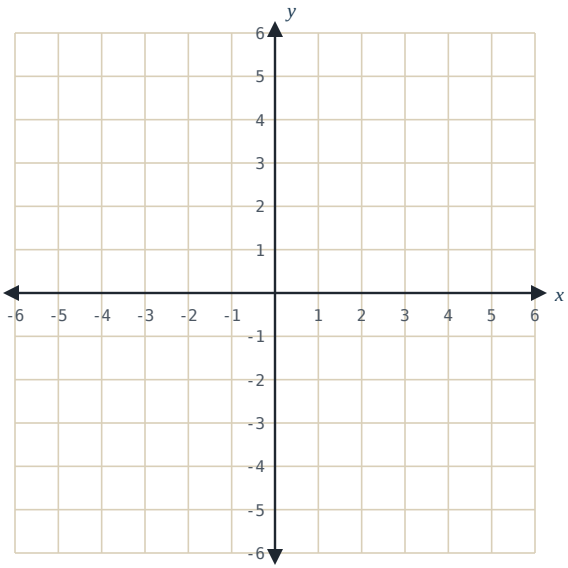
B. No solution

C. Infinite solutions

D. Cannot be determined

23. SOLVE BY GRAPHING

Graph the following system on the grid below and find the solution.



$y = x + 3$

$y = -2x + 9$

x = _____

y = _____

24. SUBSTITUTION

Solve the system by substitution. $y = 2x + 1$ $y = -x + 7$

x = _____

y = _____

25. SUBSTITUTION

Solve the system by substitution. $y = x - 4$ $3x + 2y = 7$

$x =$ _____

$y =$ _____